Data Collection and Preprocessing Phase Report

Team ID	MCA
Project Title	Smartphone Usage & Digital Well-Being Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Overview

This framework identifies data sources, assesses quality issues like missing values and duplicates, and implements targeted resolution plans to ensure accurate, robust, and reliable analysis downstream.

Phased Data Preprocessing Execution

Data Overview	Short description about dataset. Focuses on the core asset 'Smartphone_Usage.csv' containing 7,500 active user records structured across 16 analytical behavioral attributes covering demographic variables, notification frequencies, interaction volumes, everyday lifestyle factors, and categorical target addiction conditions.
Data Cleaning	Handles missing fields, standardizes unexpected data variances, eliminates duplicate structural entries, and applies system outlier correction filters to guarantee database consistency and statistical significance prior to visualization layers.
Data Transformation	Utilizes Power Query platforms to carry out filtering, multi-criteria sorting, column pivoting, attributes parsing, and generating custom calculated tracking metrics to streamline baseline dashboard performance.
Data Type Conversion	Rectifies data type formatting anomalies across fields. Guarantees true separation between floating hourly intervals, discrete trigger counts, text indicators, and binary logical structures.
Column Splitting and Merging	Executes column splitting or merging operations as needed. Segregates raw categorical ranges or age distributions into discrete operational groupings like '20-29', '30-39', and 'Under 20' for unified dimensional slices.

Data Modeling	Defines strict star-schema logical relationships across operational target matrices, binds interactive dimension boundaries, and writes standard custom measures to build cross-filtering pipelines.
Save Processed Data	Commits and saves the fully cleaned, transformed, and structured data files to local secure directory stores, establishing a pristine single source of truth for reliable and future business reporting applications.